Ryan D. Kennedy

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Objective

Electrical Engineering student with experience in embedded programming and cloud computing services, seeking a Summer 2027 internship focused on embedded system programming and design. Eager to apply and expand skills in programming, and embedded system design.

Education

Georgia Institute of Technology | Atlanta, GA *August 2026 – Present*
Bachelor of Science in Electrical Engineering Expected Graduation, May 2029
HOPE Scholarship

Kennesaw State University | Kennesaw, GA *August 2025 – May 2026*
Transfer with 68 Credit Hours, GPA 3.81/4.00
HOPE Scholarship, Dean's List

Skills

Programming: Java, Python, C, C++, JavaScript (novice), SQL, Bash, Rust (novice), Elisp, CMake, GLSL
Platforms: Linux (Ubuntu, Arch, Debian)
Hardware: Raspberry Pi, Arduino
Software: FreeCAD, Onshape, GitHub, Creality Slicer, Blender, KiCad (novice)
Languages: English (native), Spanish (beginner)

Projects

Keyboard Hero | Embedded Programming (High School) *Spring 2025*
An Individual school project that involved creating a game, game engine, and a custom controller used to play the game.

- Created an efficient rendering system using the Vulkan API to efficiently control and allocate resources on the GPU.
- Made a custom controller using CAD software and manufactured it with a 3D printer.
- Ported a Python library used to communicate with the embedded device in the controller to C++.
- Designed a flexible software architecture to allow for expansion to multiple operating systems, and hot reloadable debug builds using dynamically linked libraries.
- Architected a SQL database to store game information.
- Programmed a custom algorithm to load 3D models into program memory.
- Implemented a custom text rendering system using a generated character atlas and a formatting algorithm.

RC Light | Embedded Programming (High School) *Fall 2024*
A group school project that involved creating a network of WIFI operated servos that are used to control light switches.

- Developed a front-end application using GTK4 for interfacing with the light switch control devices.
- Implemented a multithreaded network system in the front-end application to handle multiple pending network operations at the same time.
- Programmed the network code on the control devices that are used to communicate and take commands from the controller.
- Took charge of the team and allocated tasks to achieve an efficient workflow.

Relevant Coursework

Calculus 1 & 2: Basic derivatives and integrals
Digital System Design: Boolean logic, Logic gates, Basic CMOS
Physics 2: Electromagnetism, RC & LC Circuits, Ohm's Law, Kirchoff's Law

Leadership or Activities

GT Supersonics Club | Avionics Team Member *September 2026 – Present*

- Working on avionics for the aircraft that the club is building.
- Newly acquired member that hopes to learn about avionics and contribute to the club's goal of creating a supersonic aircraft.